

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad

July 2025 – Present

Telangana, India

Master of Technology in Computer Science and Engineering

CGPA: 9.12

Indian Institute of Information Technology, Ranchi

Aug 2020 – Jun 2024

Ranchi, India

Bachelor of Technology (Hons.) in Computer Science and Engineering

CGPA: 9.05

PROJECTS

Harpy: AI Competitive Programming Toolkit | *Python, MCP Server, C++17, Subprocesses, CI/CD* 🔊

- Architected an open-standard JSON-RPC 2.0 stdio MCP server exposing 4 deterministic tools, connecting LLM coding agents (Cursor, Antigravity, Claude) to local C++17 compiler toolchains and CPH test runners.
- Engineered automated brute-force differential testing and a sandboxed multi-language runner with sub-millisecond execution benchmarking, TLE watchdog interrupts, and side-by-side terminal diffs.
- Implemented dual-channel CPH synchronization via raw HTTP sockets and offline .cph test files, distributed as a PyPI package and PyInstaller standalone binary backed by Python 3.10–3.12 CI.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE* 🔊

- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens in Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility tax and achieving 0.505 Bits-Per-Byte compression across high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT.

BoardBuddy: Real-Time Collaborative Canvas | *Next.js 14, Convex DB, Liveblocks, TypeScript* 🔊

- Architected a collaborative workspace supporting real-time multi-user syncing, freehand sketching, and vector shapes.
- Synchronized real-time canvas state and user presence across concurrent client sessions using Liveblocks and Convex DB.
- Secured organization-scoped workspaces and role permissions with Clerk authentication and a responsive TailwindCSS UI.

Peer-to-Peer Distributed File Sharing System | *C++, Sockets, Multithreading, OpenSSL* 🔊

- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers.
- Engineered out-of-order parallel chunk writing via `pwrite()` and dynamic bitfield tracking for simultaneous partial seeding.
- Architected Primary-Backup tracker failover with state synchronization, session flushing to prevent split-brain, and SHA-1 checks.

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.

AI Systems & LLMs: Model Context Protocol (MCP), Transformers (Pretraining & SFT), FlashAttention-2, RoPE, PyTorch.

Systems & Tooling: Subprocess Sandboxing, POSIX Sockets, Multithreading, Linux/Unix, Docker, Git, GitHub Actions CI.

Full-Stack & Cloud: Next.js 14, React, Node.js, Convex DB, Prisma ORM, PostgreSQL, WebSockets, REST APIs.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.